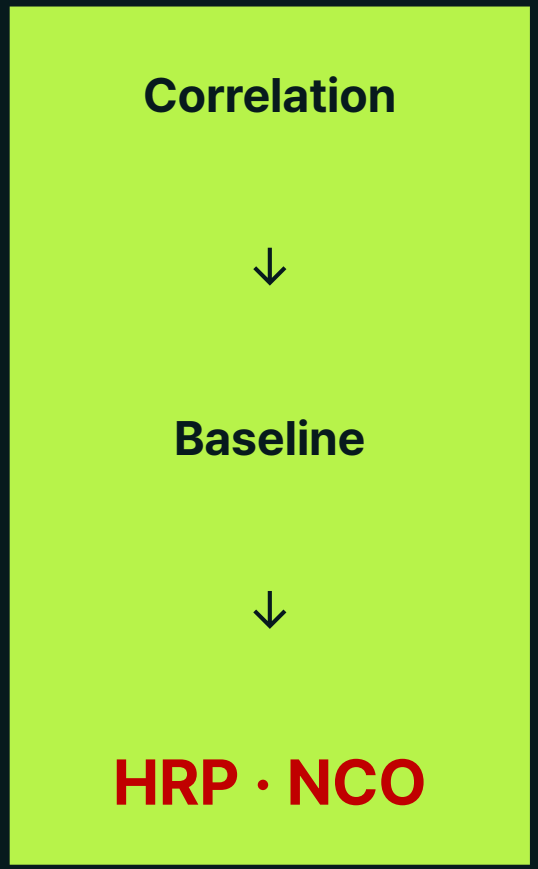
Portfolio Construction
Correlation, Allocation

Build a good mix, not just pick good stocks



Financial ML · Week 11

This week runs in four steps

From basics to core concepts, cases, then summary

01	**Basics**	**Measuring co-movement** Variance, correlation, diversification
02	**Allocation**	**Same assets, new splits** Markowitz, 1/N, inverse-variance, HRP, NCO
03	**Cases**	**Simplicity, crises** 1/N study, crisis correlation, HRP origins
04	**Summary**	**How to compare** OOS return, weight stability, cost

Distinct roles matter more than count

01

Basics

Concepts needed to read portfolio risk

Variance, covariance and correlation measure co-movement

Variance, covariance, correlation

VARIANCE

Variance

- How much one asset swings around its mean.

COVARIANCE

Covariance

- How much two assets move together.

CORRELATION

Correlation

- Covariance scaled to between −1 and 1.

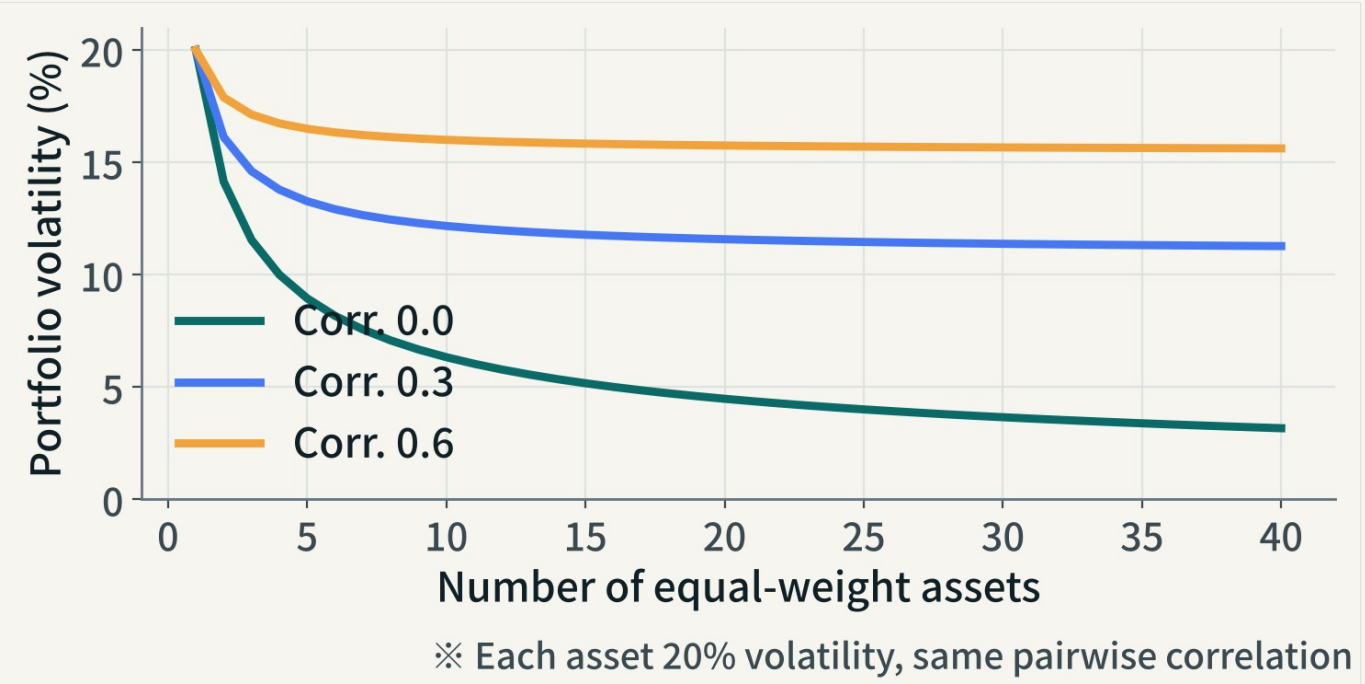
Correlation 1: always together; 0: unrelated; −1: always opposite

More assets still leave the common-correlation risk

Example: each asset 20% volatility, equal correlation, equal weight

Reading

- At zero correlation, risk keeps falling with more assets.
- With correlation, risk hits a floor.



Correlation matters more than stock count

Many similar tech stocks may not mean diversification

An optimizer blindly trusts its inputs

Garbage in, garbage out: input errors can grow in the output

INPUT

Input

- Expected returns, covariance.
- Estimated from past data, with error.

Estimates

OUTPUT

Output

- Asset weights.
- Small input changes can shift them.

Errors can amplify

So compare robust methods with a simple baseline

02

Comparing methods

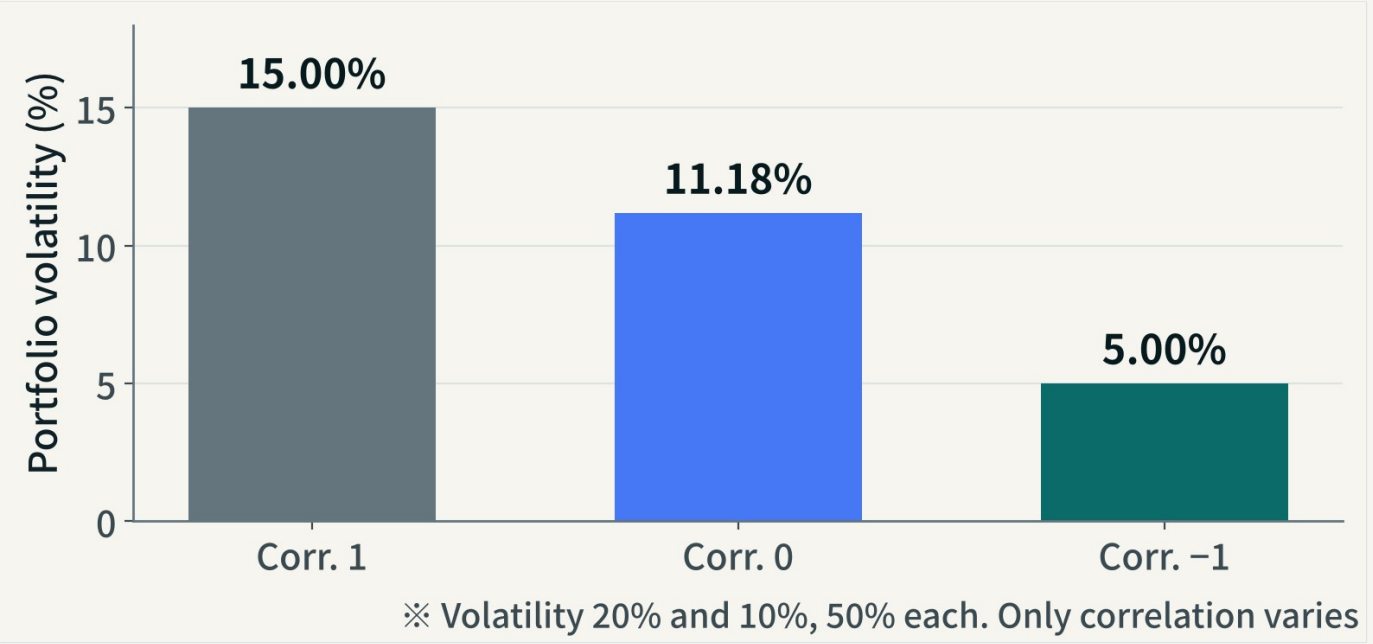
Source concepts: correlation, Markowitz, inverse-var., HRP, NCO

Picking good stocks differs from building a good mix

Two assets at 20%·10% vol., half each, varying only correlation

Team analogy

- Four students who only present well make a shaky team.
- Similar tech stocks share the same industry shocks.



Distinct roles matter more than count

Each stock's risk is unchanged, yet portfolio risk moves 15% → 5%

Diversification comes from the correlation

First two terms: each asset's risk; last term: co-movement

Two-asset variance

$$\sigma_p^2 = \underbrace{w_1^2\sigma_1^2 + w_2^2\sigma_2^2}_{\text{each asset's own variation}} + \underbrace{2w_1w_2\rho\sigma_1\sigma_2}_{\text{co-movement effect}}$$

General form

$$\sigma_p^2 = w^\top \Sigma w, \qquad \Sigma = \text{table of variances and covariances}$$

Low correlation can cut risk but does not ensure returns, and it can shift in a crisis

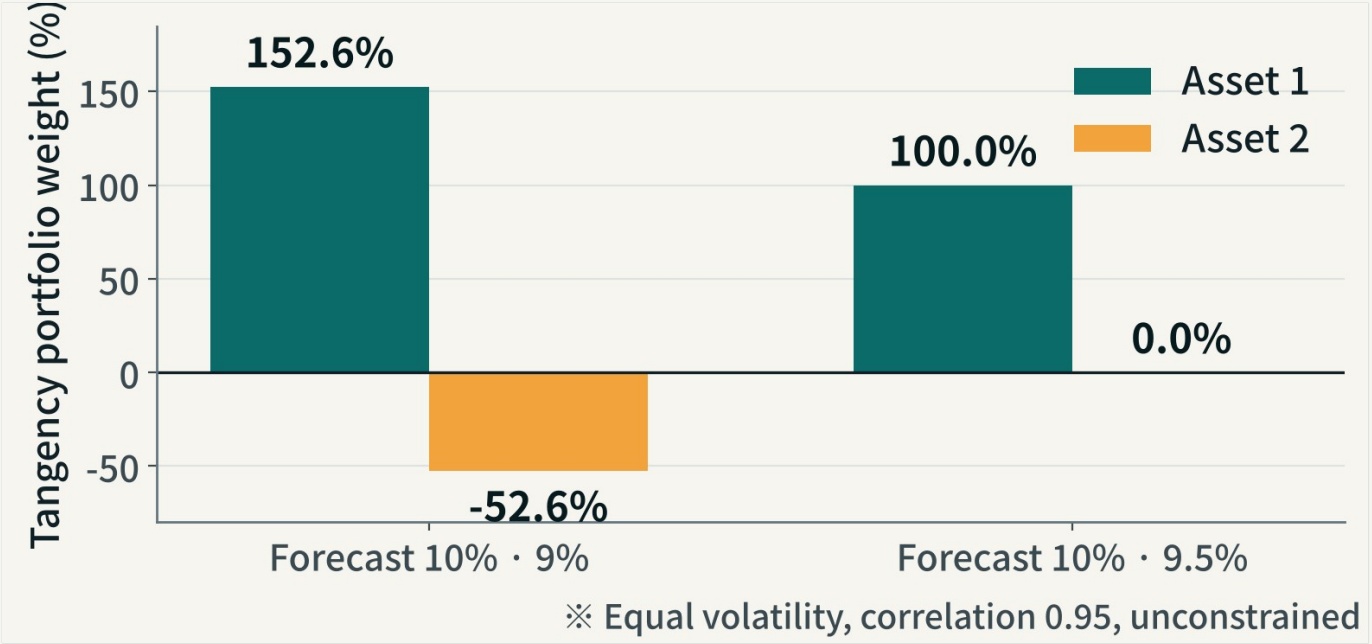
Even perfect negative correlation leaves risk if weights are off

Markowitz overreacts to small forecast differences

Two assets, equal vol., correlation 0.95, unconstrained tangency

Source p.12 example

- Forecasts 10% and 9% give 152.6% and −52.6%.
- Changing only the 2nd to 9.5% gives 100% and 0%.



Short limits, weight caps, shrinkage and regularization reduce sensitivity

Read it as ‘sensitive to input error’, not ‘always bad’

Compare simple baselines before complex ones

Same period, constraints, costs; check out-of-sample results and weight stability

Method	How it splits	What it may miss
Equal weight 1/N	Equal split over N assets	Risk size, overlap
Inverse-var. IVP	More to lower-variance assets	Asset correlations
Markowitz	Fit to goals and constraints	Sensitive to estimation error
HRP	Groups alike assets, recursive split	Clustering, window
NCO	Within, then across clusters	Inner optimization error

HRP is not always the best

Inverse-variance ≠ inverse-vol. or equal risk contribution

Inverse-variance weights for two assets at 20%·10% volatility

Inverse-variance

Inverse-variance: $w_i \propto \frac{1}{\sigma_i^2}$

Inverse-variance → 20% · 80%

- Proportional to 1/0.04 and 1/0.01.

Inverse-vol. → other

- Uses 1/volatility, so weights differ.

Equal risk contrib.

- Not the same as inverse-variance.

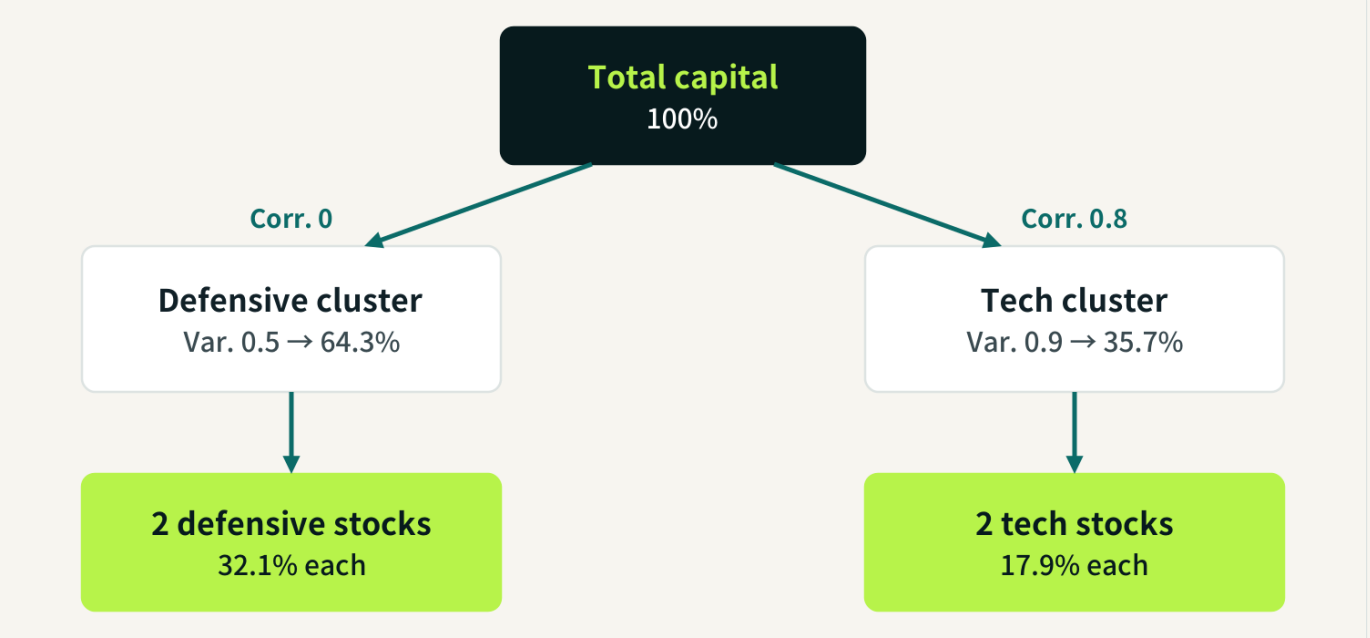
Similar names, different weighting rules

HRP looks at the asset family tree first

HRP (Hierarchical Risk Parity) — two given clusters (source p.18), each stock variance 1

Four steps

- Make distances from ρ.
- Group similar assets in a tree.
- Fix the tree order.
- Split capital by sub-cluster risk.



No inverse covariance matrix needed

Two co-moving tech stocks are not two separate bets

50:50 within clusters; defensive cluster variance 0.5, tech 0.9

Across clusters (∝ 1/cluster variance)

$$w_{\text{def}} = \frac{1/0.5}{1/0.5 + 1/0.9} = 64.3\%, \qquad w_{\text{tech}} = 35.7\%$$

Defensive (ρ 0)

- 64.3% split, about 32.1% each.

Tech (ρ 0.8)

- 35.7% split, about 17.9% each.

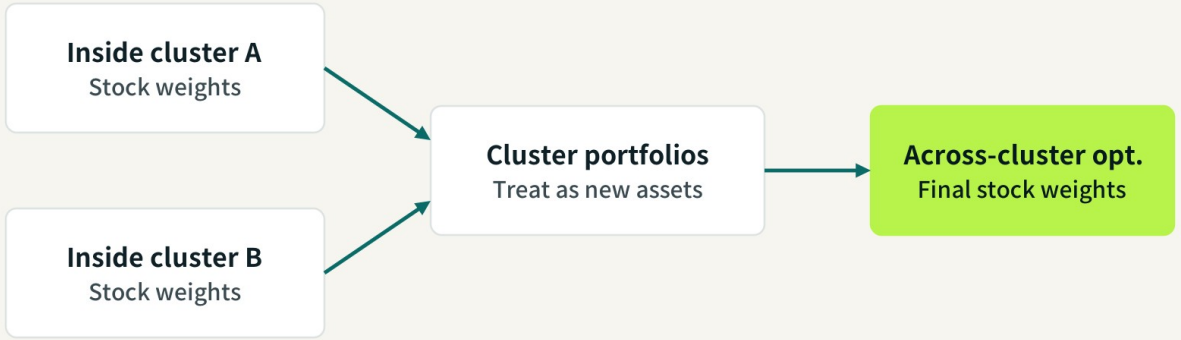
Caution

- An example, not a recommendation.

Despite its name, HRP does not equalize every asset's risk contribution

NCO builds small teams first, then optimizes across teams

NCO (Nested Clustered Optimization) — within cluster → representative → across clusters



40% within cluster A, 30% to cluster A

$$\text{final weight} = \underbrace{0.4}_{\text{within cluster}} \times \underbrace{0.3}_{\text{cluster allocation}} = 12\%$$

It can use cross-cluster correlation and is not the HRP algorithm

※ Read the steps to check whether the source's HRP-style code matches true HRP

03

Lessons from cases

What research and markets have shown

Many optimizers failed to beat simple 1/N consistently

DeMiguel·Garlappi·Uppal(2009, Review of Financial Studies)

What was tested

- 14 mean-variance models vs. 1/N, out-of-sample.

Results

- No model beat 1/N consistently.

Why

- Estimation error offset optimization gains.

Markowitz reportedly split his own retirement savings half stocks, half bonds

A complex method must beat a simple baseline to be worth it

In crises correlations rise and diversification shrinks

A mix built on calm-period correlations may be less diversified in a crisis

LTCM (1998)

- Trades thought different lost together in one liquidity shock.

2008 crisis

- Many risky assets fell at once.

Lesson

- Check weights and risk over crisis periods separately.

Low correlation cuts risk but is never guaranteed to last

HRP was proposed to avoid unstable inverse covariance

López de Prado (2016, Journal of Portfolio Management): four steps

Distance via correlation → **Cluster similar assets** → **Order assets by tree** → **Allocate by cluster risk**

Strengths

- No matrix inversion, so less sensitive to input error.

Limits

- Results vary with clustering method and window.

Judge by implementation and OOS results, not name

04

Summary

Close the week with key Q&A

Choose a mix by correlation structure and OOS tests

One question, one answer for this week

Check question	Answer
Are many stocks enough to diversify?	**Not if they move together in similar shocks.**
Is HRP always best?	**No. Compare OOS with the same period, limits and costs.**
Inverse-var. = inverse-vol.?	**No. One uses 1/variance, the other 1/volatility.**
Why is Markowitz fragile?	**It overtrusts small forecast gaps and swings weights.**

Compare return, risk, turnover and costs in one table

PICK ROLES, NOT NAMES.

START WITH 1/N.

BEAT IT OUT OF SAMPLE.

Go complex only when it
beats a simple baseline.